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Top Skills

Cloud Application Development
Knex.js
Twilio

Certifications

Node.js
Discovering Personality
Data Cleaning and Preprocessing With Pandas
Advance Javascript
Deep Learning

Vibek Dutta

Node.js | NestJS | React | Typescript | Go | AWS | Kubernetes | Terraform
Greater Toronto Area, Canada

Summary

Most of the bugs I've actually enjoyed chasing weren't in the code that broke. They were two or three services upstream from it, hiding behind a stack trace that pointed somewhere completely innocent.I treat misleading errors as clues, tracing logs backward to find where reality broke the code's assumptions. I've worked on syncing state across distributed systems, scaling APIs for massive traffic, and building resilient frontends designed to survive unpredictable user behavior. Slow queries and lock contention show up in there too, they're just one version of a more general pattern: something assumed a well behaved world, and got a messy one instead.Outside of work, I write about this kind of thing and dig through open source projects, mostly to see how someone else solved the same headache differently.

Experience

KPM Power Inc
Full Stack Engineer
April 2026 - Present (7 months)
Etobicoke, Ontario, Canada

Resso.ai - Career and Practice Platform
Cloud Engineer
July 2025 - March 2026 (9 months)
Ontario, Canada

• Built an AI recommendation platform using Python, DeepSeek LLM, AstraDB vector search, and RAG to deliver personalized course recommendations, with Redis-based KV caching to reuse retrieved context and reduce unnecessary LLM processing.
• Integrated React applications with Python AI services, supporting real-time LLM streaming through Server-Sent Events (SSE), session caching, and reliable error handling.

• Created CI/CD pipelines with GitHub Actions for React, Node.js, and Python applications, automating tests, code quality checks, and deployments across environments.
• Deployed and managed backend and AI services using Docker, Kubernetes, and Helm, with rolling updates, health checks, and multi-stage builds for consistent releases.
• Built event-driven workflows with API Gateway, AWS Lambda, SQS and EventBridge to automate notifications and trigger AI-powered analysis based on student learning activity.
• Monitored AI applications and infrastructure using Grafana and AWS CloudWatch, tracking LLM response times, Redis cache behavior, resource usage, and service health.
• Implemented secure access using JWT, RBAC, and MFA for distributed services, APIs, and AI tooling.
• Automated AI model deployments, Redis cache warmup, and log maintenance using Python and Bash to improve service readiness before live traffic.

Snappymob

Software Engineer

June 2022 - February 2024 (1 year 9 months)

WP. Kuala Lumpur, Federal Territory of Kuala Lumpur, Malaysia

• Rebuilt dashboard in React 18, TypeScript, and Vite. Replaced polling with Server-Sent Events, used TanStack Query for server state management, and chose Zustand over Redux for a lighter state management approach. Resolved rendering bottlenecks using React.memo, useMemo, and react-window for large datasets.
• Structured Turborepo monorepo for CI speed vs. Nx's plugin breadth; remote caching eliminated redundant NestJS builds, cutting pipeline time without Nx overhead.
• Designed MySQL transaction schema with normalized tables, foreign keys, and ACID boundaries. Abstracted data access behind typed repositories; swapped in read-replica for reporting without API changes.
• Implemented CQRS with RabbitMQ for write-side commands (payments, refunds), Kafka for query-side eventual consistency. Used Saga pattern with DLQs to coordinate workflows without distributed transactions.
• Built cron reconciliation via AWS EventBridge: validated MySQL records against payment provider APIs and wrote discrepancies to SQS. Idempotent re-run semantics prevented double-counting on retries.

• Built data ingestion pipelines with AWS Step Functions, Kinesis Firehose, Lambda, S3, and Redshift. Orchestrated workflows with Airflow and Docker, and used Kafka Streams for real-time aggregation and event processing.
• Architected multi-stage payment onboarding wizard using React Hook Form with useFieldArray for dynamic line items, useWatch for real-time cross-field validation, and zodResolver for schema-based validation. Implemented React Context with useReducer for step state management, React.memo with custom areEqual for performance, and framer-motion for animated transitions.

RedSoft Solutions Pvt. Ltd.
Blockchain Developer
November 2022 - April 2023 (6 months)
Ahmedabad, Gujarat, India

• Developed an NFT marketplace on Polygon using Solidity and Hardhat, implementing ERC721 for unique digital assets and ERC1155 for tokenized perks and customer rewards.
• Built a Node.js and TypeScript backend with Alchemy and Ethers.js for wallet authentication, transaction monitoring, smart contract event listeners, and queue-based processing for on-chain events.
• Integrated Chainlink Price Feeds for NFT valuation support and Chainlink VRF for fair randomization in limited-edition drops.
• Designed MongoDB schemas for off-chain NFT metadata, using change streams for real-time updates and aggregation pipelines for marketplace analytics.
• Built RESTful APIs for the React frontend and third-party integrations, with JWT authentication, rate limiting, and webhooks for transaction notifications.
• Used ERC1155 tokens to support exclusive customer perks such as hotel discounts, rewards, and automated redemption workflows.

KloudMate
Fullstack Web Developer
July 2021 - May 2022 (11 months)
Bengaluru, Karnataka, India

• Built React.js frontend using Hooks, Material-UI, Apollo Client, and WebSockets, with Sentry for error tracking, AWS Cognito for JWT authentication, and AWS Amplify for CI/CD deployments.
• Optimized Apollo Client using InMemoryCache, optimistic UI updates, and fetch policies to improve responsiveness while keeping data fresh.
• Developed an AWS Step Functions visualizer using React and D3.js, parsing ASL JSON to render state workflows with AWS X-Ray trace integration.

• Built Node.js and TypeScript backend services with Apollo Server, GraphQL resolvers, DataLoader batching, Jest tests, and Elasticsearch Query DSL for log search.
• Created PostgreSQL views in Hasura for denormalized reads and configured Event Triggers to invoke AWS Lambda on database changes.
• Automated scheduled backend workflows using AWS EventBridge Scheduler, Lambda, SQS, and RDS for data syncing and downstream processing.
• Built observability workflows using CloudWatch Logs, Metrics, Alarms, and custom dashboards for service health monitoring.
• Architected log processing pipeline with CloudWatch, Kinesis, Elasticsearch, S3, Glue, and Athena for real-time search, backup storage, and SQL-based log analysis.
• Implemented CI/CD using Jenkins and later migrated deployments to AWS CodePipeline, CodeBuild, and CodeDeploy for blue-green ECS releases.
• Worked hands-on with AWS services including Lambda, S3, EC2, Auto Scaling Groups, Application Load Balancer, API Gateway, EventBridge, SQS, RDS, Cognito, and Kinesis.

Cyborn Ltd
Fullstack Web Developer
August 2020 - June 2021 (11 months)
Bangalore Urban, Karnataka, India

• Built responsive monitoring dashboards using Angular, TypeScript, RxJS observables, and custom operators to display real-time system metrics in a clear and actionable format.
• Developed RESTful APIs with Node.js and Express, serving JSON data to the frontend with validation middleware, error handling, and structured logging.
• Created an automated billing system with Stripe API integration, handling payment webhooks and retry logic for failed transactions.
• Migrated data from Neo4j to AWS RDS PostgreSQL using Python transformation scripts, preserving relationships while adapting graph data to relational query patterns.
• Contributed to CI/CD workflows with Jenkins, using Git-based collaboration, code reviews, and shared quality standards across the team.

zekeLabs
Deep Learning Intern
May 2018 - August 2018 (4 months)
Bangalore Urban, Karnataka, India

• Built data preprocessing pipelines using Pandas and NumPy, applying PCA for dimensionality reduction and VIF analysis to handle multicollinearity in high-feature datasets.
• Orchestrated ETL workflows with AWS Step Functions, Glue, Athena, and S3 data lakes, including schema crawling from unstructured data.
• Developed deep learning models with PyTorch, including CNNs, LSTMs, and transfer learning with ResNet and VGG, using batch normalization, dropout, Adam, SGD, and TensorBoard tracking.
• Evaluated model performance using cross-validation, F1-score, and ROC-AUC, while reducing overfitting with L2 regularization and early stopping.
• Containerized ML workloads with Docker, including TensorFlow, PyTorch, CUDA, cuDNN, Pandas preprocessing, and Express.js REST APIs for predictions.
• Deployed multi-container ML services with Docker Compose, using Redis for feature store caching and PostgreSQL for prediction logging.

Education

Conestoga College
Postgraduate Degree, Reporting Systems and Database Development · (September 2024 - April 2025)

Conestoga College
Postgraduate Degree, Big Data Solutions Architecture · (September 2023 - April 2024)

Rajasthan Technical University
Bachelor of Technology, Computer Science And Engineering · (2016 - 2020)

Vivekananda Kendra Vidyalaya
High School Diploma, Science · (2013 - 2015)